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(54) SYSTEM AND METHOD FOR DETERMINING A DRIVER RETIREMENT SCORE

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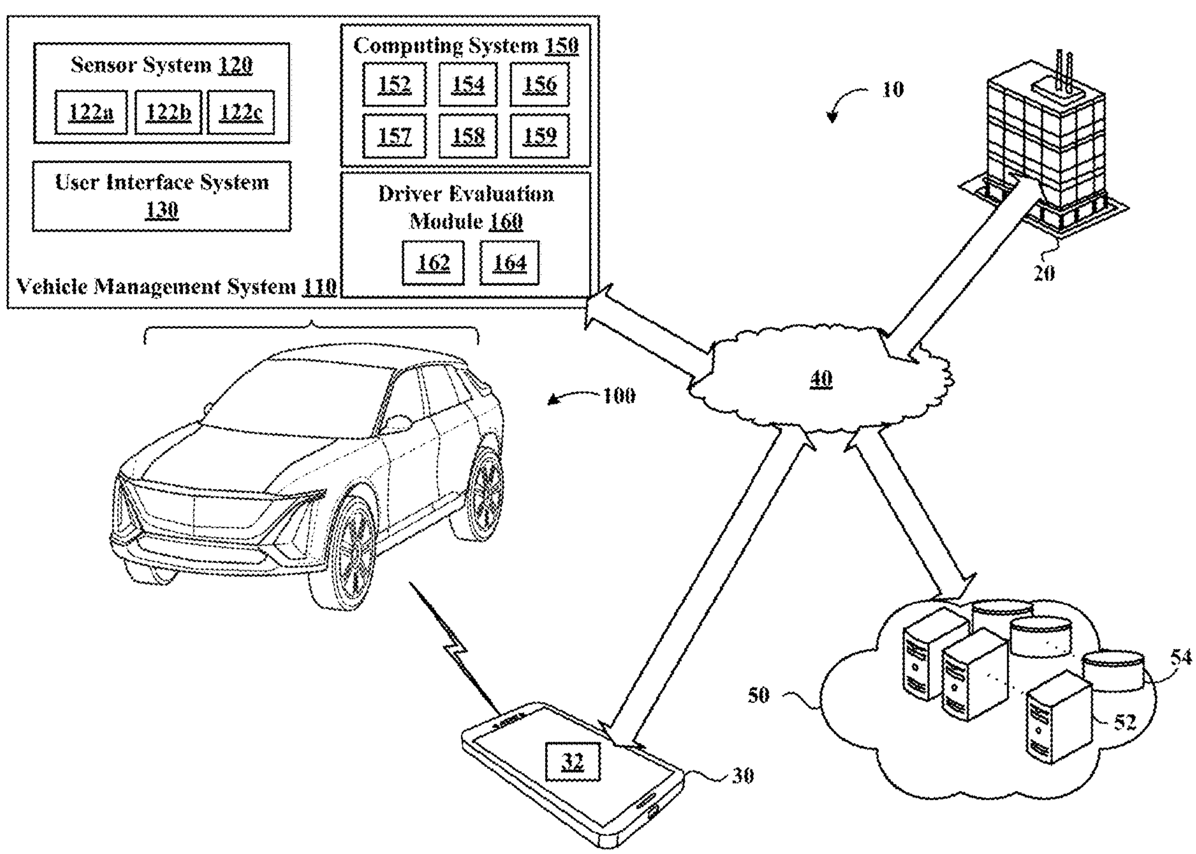
(57) ABSTRACT

A computer-implemented method that, when executed by data processing hardware, causes the data processing hardware to perform operations comprising gathering vehicle and driver data, receiving one or more scoring inputs based on the vehicle and driver data, calculating a driver retirement score based on the one or more scoring inputs, assessing a driver of a vehicle with a performance category based on the driver retirement score, and notifying the driver of the performance category.



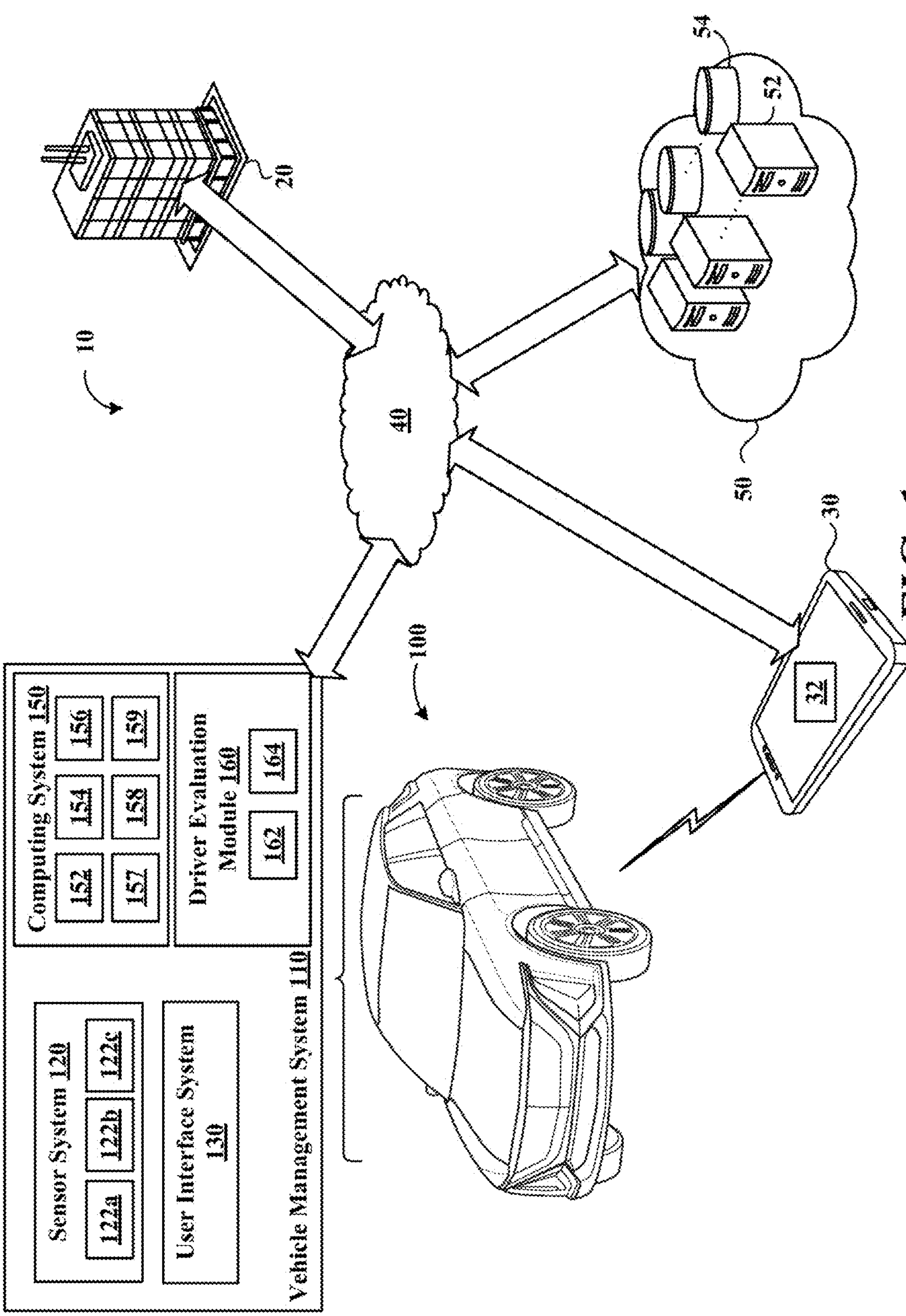

10
20
40
50
52
54
30
32
100
Computing System 150
152
154
156
157
158
159
Driver Evaluation Module 160
162
164
Sensor System 120
122a
122b
122c
User Interface System 130
Vehicle Management System 110

FIG. 1

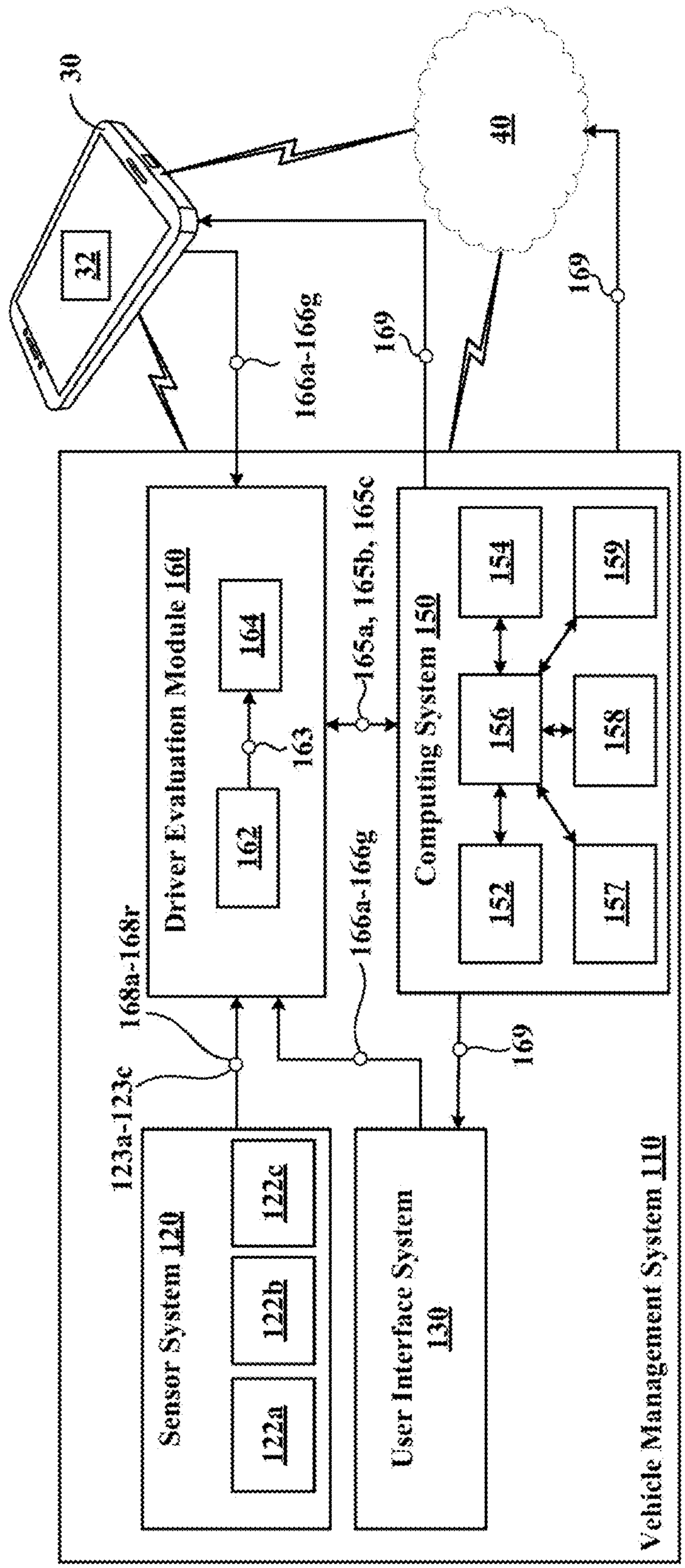

30
32
40
166a-166g
169
169
Driver Evaluation Module 160
164
162
163
165a, 165b, 165c
Computing System 150
154
159
156
158
152
157
168a-168r
123a-123c
166a-166g
169
Sensor System 120
122a
122b
122c
User Interface System 130
Vehicle Management System 110

FIG. 2

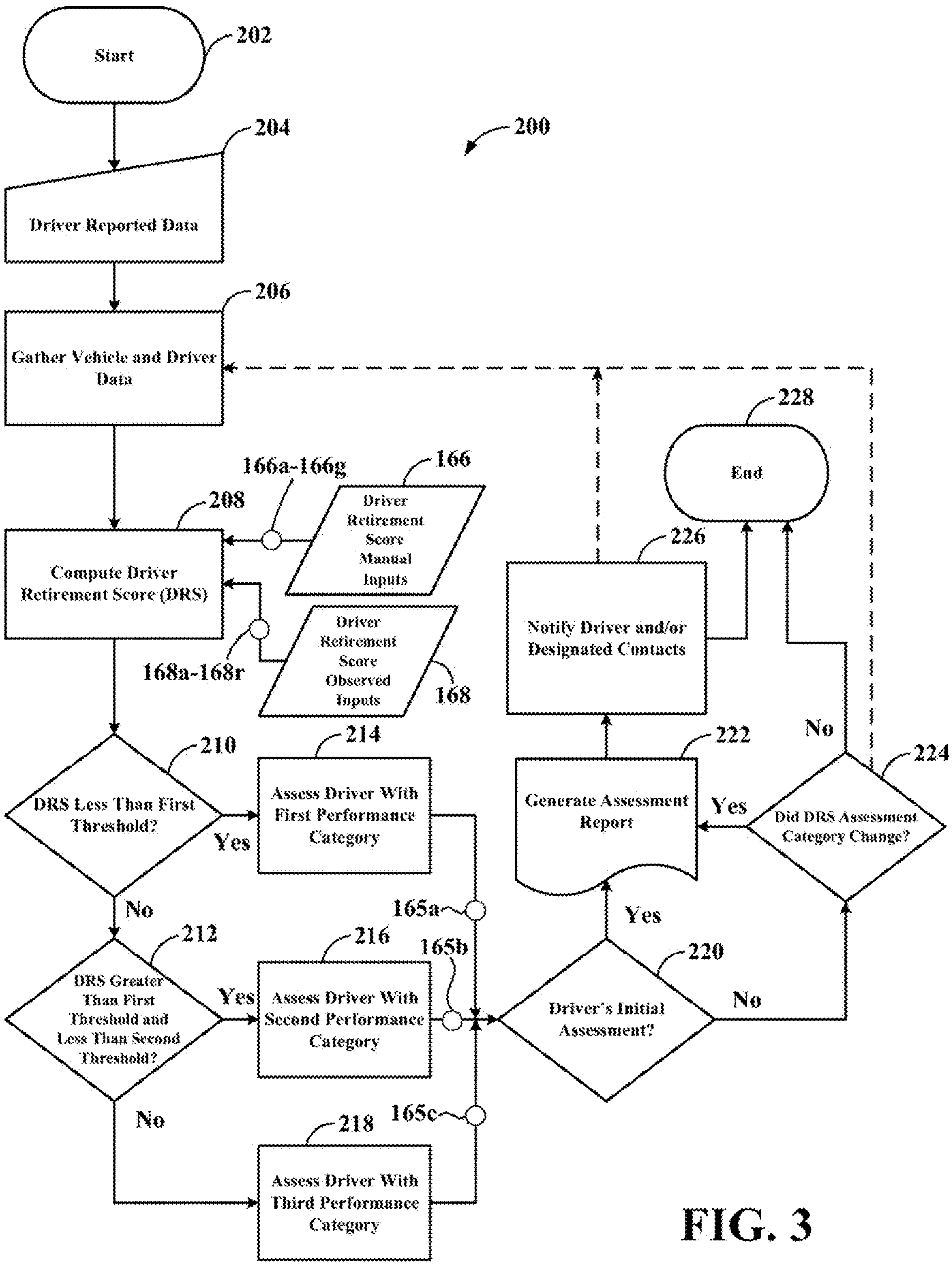

200
Start
202
Driver Reported Data
204
Gather Vehicle and Driver Data
206
Compute Driver Retirement Score (DRS)
208
Driver Retirement Score Manual Inputs
166
166a-166g
Driver Retirement Score Observed Inputs
168
168a-168r
DRS Less Than First Threshold?
210
Yes
No
Assess Driver With First Performance Category
214
DRS Greater Than First Threshold and Less Than Second Threshold?
212
Yes
No
Assess Driver With Second Performance Category
216
Assess Driver With Third Performance Category
218
165a
165b
165c
Driver's Initial Assessment?
220
Yes
No
Generate Assessment Report
222
Notify Driver and/or Designated Contacts
226
End
228
Did DRS Assessment Category Change?
224
Yes
No

FIG. 3

SYSTEM AND METHOD FOR DETERMINING A DRIVER RETIREMENT SCORE

INTRODUCTION

[0001] The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0002] The present disclosure relates generally to a system and method for determining a driver retirement score, and more particularly, assessing performance of an older driver.

[0003] In general, aging can be associated with physical and psychological declines. These declines can make driving a vehicle difficult or frustrating for some older individuals. However, deciding to retire from driving a vehicle can be difficult as driving can provide a sense of freedom and/or independence that many rely on to perform day-to-day activities. Some systems and methods may evaluate performance of a driver, but they fail to provide objective assessments to the driver for further evaluation. These shortcomings, among others, are addressed by aspects of the present disclosure.

SUMMARY

[0004] In one configuration, a computer-implemented method that, when executed by data processing hardware, causes the data processing hardware to perform operations is provided. The operations include gathering vehicle and driver data, receiving one or more scoring inputs based on the vehicle and driver data, calculating a driver retirement score based on the one or more scoring inputs, assessing a driver of a vehicle with a performance category based on the driver retirement score, and notifying the driver of the performance category.

[0005] The method may include one or more of the following optional aspects or steps. For example, the one or more scoring inputs may include driver retirement score manual inputs and driver retirement score observed inputs.

[0006] According to at least one aspect, the one or more scoring inputs may include driver retirement score manual inputs that increase or decrease the driver retirement score and include at least one of the following: a medical condition quantity, a medical condition type, a complaint status, a path following status, a road sign compliance status, a driver confidence status, and a driver physical ability status.

[0007] In at least one example, the one or more scoring inputs may include driver retirement score observed inputs that increase or decrease the driver retirement score and include at least one of the following: a driver fatigue status, a driver distraction status, a speed-limit following status, a driver squinting status, a honking status, a turn signal utilization status, a vehicle straddling status, a vehicle drifting status, a driver reaction time status, a vehicle following distance status, a driving record status, a driver posture status, a vehicle speed control status, an accelerator and brake status, a navigation deviation status, a vehicle control status, and a dangerous behavior status.

[0008] According to another aspect, the performance category may include a first performance category, a second performance category, and a third performance category. The first performance category may be associated with acceptable driver performance, the second performance category may be associated with questionable driver performance, and the third performance category may be associated with unacceptable driver performance. The first performance category may be associated with acceptable physical and psychological ability for driving, the second performance category may be associated with questionable physical and psychological ability for driving, and the third performance category may be associated with unacceptable physical and psychological ability for driving.

[0009] According to another example, assessing the driver of a vehicle with a performance category based on the driver retirement score may include assessing a first performance category if the driver retirement score is less than a first threshold, assessing a second performance category if the driver retirement score is greater than the first threshold but less than a second threshold, and assessing a third performance category if the driver retirement score is greater than both the first and second thresholds.

[0010] In at least one aspect, the method further includes generating a report based on the one or more scoring inputs and the performance category and providing personalized recommendations for the driver via a user interface system or a mobile device.

[0011] In at least another aspect, the method further includes determining whether the performance of the driver is improving or degrading and providing personalized recommendation in a report based on any improvement or degradation of performance.

[0012] In at least one example, notifying the driver may further include notifying a designated contact of the driver.

[0013] In another configuration, a vehicle management system is provided and includes a

[0014] sensor system comprising one or more sensor subsystems, data processing hardware, and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that, when executed on the data processing hardware, cause the data processing hardware to perform operations. The operations include gathering vehicle and driver data via the sensor system and via one or more manual inputs from a driver, receiving one or more scoring inputs based on the vehicle and driver data, calculating a driver retirement score based on the one or more scoring inputs, assessing a driver of a vehicle with a performance category based on the driver retirement score, generating a report with personalized recommendations for the driver based on the one or more inputs and the performance category, and notifying the driver and one or more designated contacts of the report.

[0015] According to at least one configuration, the one or more scoring inputs may include driver retirement score manual inputs and driver retirement score observed inputs.

[0016] According to at least one aspect, the one or more scoring inputs may include driver retirement score manual inputs that increase or decrease the driver retirement score and include at least one of the following: a medical condition quantity, a medical condition type, a complaint status, a path following status, a road sign compliance status, a driver confidence status, and a driver physical ability status.

[0017] In at least one example, the one or more scoring inputs may include driver retirement score observed inputs that increase or decrease the driver retirement score and include at least one of the following: a driver fatigue status, a driver distraction status, a speed-limit following status, a driver squinting status, a honking status, a turn signal utilization status, a vehicle straddling status, a vehicle drifting status, a driver reaction time status, a vehicle following distance status, a driving record status, a driver posture status, a vehicle speed control status, an accelerator and brake status, a navigation deviation status, a vehicle control status, and a dangerous behavior status.

[0018] According to another aspect, the performance category may include a first performance category, a second performance category, and a third performance category. The first performance category may be associated with acceptable driver performance, the second performance category may be associated with questionable driver performance, and the third performance category may be associated with unacceptable driver performance. The first performance category may be associated with acceptable physical and psychological ability for driving, the second performance category may be associated with questionable physical and psychological ability for driving, and the third performance category may be associated with unacceptable physical and psychological ability for driving.

[0019] According to another example, assessing the driver of a vehicle with a performance category based on the driver retirement score may include assessing a first performance category if the driver retirement score is less than a first threshold, assessing a second performance category if the driver retirement score is greater than the first threshold but less than a second threshold, and assessing a third performance category if the driver retirement score is greater than both the first and second thresholds.

[0020] In another aspect, the method further comprises determining whether the performance of the driver is improving or degrading and providing personalized recommendation in the report based on any improvement or degradation of performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The drawings described herein are for illustrative purposes only of selected configurations and are not intended to limit the scope of the present disclosure.

[0022] FIG. **1** is a schematic diagram of a vehicle environment including a vehicle management system according to the principles of the present disclosure;

[0023] FIG. **2** is an enlarged schematic diagram showing an example of the vehicle management system according to the principles of the present disclosure; and

[0024] FIG. **3** is a flow diagram showing operations of the vehicle management system of FIG. **2**.

[0025] Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0026] Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

[0027] The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles "a," "an," and "the" may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms "comprises," "comprising," "including," and "having," are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

[0028] When an element or layer is referred to as being "on," "engaged to," "connected to," "attached to," or "coupled to" another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," "directly attached to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0029] The terms "first," "second," "third," etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

[0030] In this application, including the definitions below, the term "module" may be replaced with the term "circuit." The term "module" may refer to, be part of, or include an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor (shared, dedicated, or group) that executes code; memory (shared, dedicated, or group) that stores code executed by a processor; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

[0031] The term "code," as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, and/or objects. The term "shared processor" encompasses a single processor that executes some or all code from multiple modules. The term "group processor" encompasses a processor that, in combination with additional processors, executes some or all code from one or more modules. The term "shared memory" encompasses a single memory that stores some or all code from multiple modules. The term "group memory" encompasses a memory that, in combination with additional memories, stores some or all code from one or more modules. The term "memory" may be a subset of the term "computer-readable medium." The term "computer-readable medium" does not encompass transitory electrical and electromagnetic signals propagating through a medium, and may therefore be considered tangible and non-transitory memory. Non-limiting examples of a non-transitory memory include a tangible computer readable medium including a nonvolatile memory, magnetic storage, and optical storage.

[0032] The apparatuses and methods described in this application may be partially or fully implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on at least one non-transitory tangible computer readable medium. The computer programs may also include and/or rely on stored data.

[0033] A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an "application," an "app," or a "program." Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

[0034] The non-transitory memory may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by a computing device. The non-transitory memory may be volatile and/or non-volatile addressable semiconductor memory. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0035] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" and "computer-readable medium" refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0036] Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICS (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0037] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0038] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0039] Referring to FIG. **1**, an example vehicle operating environment **10** is provided for illustration of the principles of the present disclosure. The vehicle operating environment **10** includes a vehicle **100**, a vehicle service center **20**, and a mobile device **30**. For the sake of illustration, the vehicle operating environment **10** is shown as including a single vehicle service center **20**. However, in other examples, the vehicle operating environment **10** may include a plurality of vehicle service centers **20** in communication over a network **40** (e.g., the Internet, cellular networks).

[0040] With reference to FIGS. **1** and **2**, the vehicle **100** includes a vehicle management system **110** including a sensor system **120**, a vehicle user interface system **130**, a computing system **150**, and a driver evaluation module **160**. The vehicle management system **110** may be configured for gathering information from the vehicle **100** and/or from a driver (i.e., an operator) to evaluate, calculate, or otherwise provide a driver retirement score **163**. The driver retirement score **163** may be desirable so that the driver may be assessed with a performance category **165** and one or more recommendations may be provided to the driver and/or a designated contact of the driver (e.g., a family member, friend, spouse, etc.).

[0041] While the vehicle **100** maneuvers about the environment **10**, the sensor system **120** includes various sensor subsystems **122**, **122*a*-122*c*** configured to gather sensor data **123**, **123*a*-123*c*** relating to characteristics of the environment **10** and/or status of the vehicle **100**. For instance, the sensor subsystems **122** include a vehicle exterior sensor subsystem **122*a*** configured to measure or obtain external environmental data **123*a***, such as weather or surrounding objects (e.g., vehicles, pedestrians). The vehicle exterior sensor subsystem **122*a*** can include one or more of an RGB camera, an infrared camera, a thermal camera, a radar, and/or an external microphone, for example. The sensor subsystems **122** also include an interior sensor subsystem **122*b*** configured to measure interior environmental data **123*b***, such as driver facial expressions or posture. The interior sensor subsystem **122*b*** can include one or more of biosensors, a driver monitor system camera, an RGB camera, an infrared camera, a thermal camera, a radar, and/or an internal microphone, for example. A vehicle status subsystem **122*c*** configured to measure or obtain vehicle operating data **123*c***, such as vehicle location and operating parameters such a wheel speed or lateral acceleration. The vehicle status subsystem **122*c*** can include one or more wheel speed sensors and an inertial measurement unit, for example.

[0042] As the sensor system **120** gathers the sensor data **123**, the computing system **150** is configured to store, process, and/or communicate the sensor data **123** within the vehicle operating environment **10**. In order to perform computing tasks related to the sensor data **123**, the computing system **150** of the vehicle **100** includes data processing hardware **152** and memory hardware **154**. The data processing hardware **152** is configured to execute instructions stored in the memory hardware **154** to perform computing tasks related to operation and management of the vehicle **100**. Generally speaking, the computing system **150** refers to one or more locations of data processing hardware **152** and/or memory hardware **154**.

[0043] The data processing hardware **152** may be embodied as a discrete microprocessor, an application specific integrated circuit (ASIC), or a dedicated control module. The vehicle **100** may offer centralized vehicle control via a central processing unit (CPU) **156** that is coupled to memory hardware **154** each of which may take on the form of a CD-ROM, magnetic disk, IC device, semiconductor memory (e.g., various types of RAM or ROM), etc., and a real-time clock (RTC). Long range vehicle communication capabilities, such as Global Network Satellite System (GNSS), with remote, off-board networked devices may be provided via one or more or all of a cellular chipset/component, a navigation and location chipset/component (e.g., global positioning system (GPS)), or a wireless modem, all of which are collectively represented at **157**. Close-range wireless connectivity may be provided via a short-range wireless communication device **158** (e.g., a Bluetooth® unit or near field communications (NFC) transceiver), and/or a dual antenna **159**.

[0044] In some examples, the computing system **150** is a local system located on the vehicle **100**. When located on the vehicle **100**, the computing system **150** may be centralized (i.e., in a single location/area on the vehicle **100**), decentralized (i.e., located at various locations about the vehicle **100**), or a hybrid combination of both (e.g., with a majority of centralized hardware and a minority of decentralized hardware). To illustrate some differences, a decentralized computing system **150** may allow processing to occur at an activity location while a centralized computing system **150** may allow for a central processing hub that communicates to systems located at various positions on the vehicle **100**.

[0045] Additionally or alternatively, the computing system **150** includes computing resources that are located remotely from the vehicle **100**. For instance, the computing system **150** may communicate via the network **40** with a remote vehicle computing system **50** (e.g., a remote computer/server or a cloud-based environment). Much like the computing system **150**, the remote vehicle computing system **50** includes remote computing resources such as remote data processing hardware **52** and remote memory hardware **54**. Here, sensor data **123** or other processed data (e.g., data processed locally by the computing system **150**) may be stored in the remote vehicle computing system **50** and may be accessible to the computing system **150**. In some examples, the computing system **150** is configured to utilize the remote resources **52**, **54** as extensions of the computing resources **152**, **154** such that resources of the computing system **150** may reside on resources of the remote vehicle computing system **50**.

[0046] With continued reference to FIGS. **1** and **2**, the vehicle management system **110** includes the driver evaluation module **160**, which is executed at the local computing system **150** and/or the remote vehicle computing system **50** in the manner previously described. The driver evaluation module **160** can include a driver retirement score calculator **162** and a driver assessment module **164**. The driver retirement score calculator **162** may be configured to receive the sensor data **123**, **123*a*-123*c*** from the vehicle **100** and/or from the driver to determine the driver retirement score **163**. The driver assessment module **164** may be configured to receive the driver retirement score **163** from the driver retirement score calculator **162** and assess the driver with one or more of the performance categories **165**. The performance categories **165** may include a first performance category **165*a***, a second performance category **165*b***, and a third performance category **165*c***. The first performance category **165*a*** can be associated with acceptable driver performance, the second category **165*b*** can be associated with questionable driver performance, and the third category **165*c*** can be associated with unacceptable driver performance. The performance categories **165***a*-**165***c* may also be respectively associated with acceptable physical and psychological abilities of the driver, questionable physical and psychological abilities of the driver, and unacceptable physical and psychological abilities of the driver, for example.

[0047] The vehicle management system **110** also includes the vehicle user interface system **130** which provides a medium for the driver to communicate with the vehicle **100** and/or receive information from the vehicle **100**. For instance, the vehicle user interface system **130** may include one or more buttons or levers such as a turn signal indicator, climate controls, etc. The vehicle user interface system **130** can also include a touchscreen display (e.g., an infotainment system) that can be configured to display personalized recommendations for the driver, as will be discussed in more detail below.

[0048] With reference again to FIG. **1**, the mobile device **30** may include a smartphone, tablet, or another mobile computer capable of communicating with the vehicle **100** via long range communication devices **157** or short range communication devices **158**. The mobile device **30** can be configured so that the driver or operator can provide one or more inputs to the vehicle **100** and, more particularly, to the driver evaluation module **160**. According to at least one aspect of the present disclosure, the mobile device **30** may include an original equipment manufacturer application or mobile application **32** which provides a medium so that the driver can communicate with the vehicle **100**. The mobile application **32** may be configured so that the driver can answer one or more questions and voluntarily provide information that may be desirable for the driver evaluation module **160** to calculate the driver retirement score **163**.

[0049] Referring to FIG. **3**, a method **200** for determining the driver retirement score **163** and assessing the driver with one or more of the performance categories **165** is provided. At **202**, the method **200** is initiated. In practical terms, the method **200** may be initiated upon powering-up of the vehicle **100** by the vehicle operator. Upon initiation, but prior to being assessed, the driver evaluation module **160** may require affirmation, permission, and/or acknowledgment by the driver, which may be provided via the vehicle user interface system **130** and/or the mobile device **30**. In other words, driver evaluation module **160** may require the driver to "opt-in" prior to being assessed.

[0050] At **204**, the driver may optionally provide information (i.e., data) via the mobile application **32** and/or the vehicle user interface system **130** concerning age, health history, driving record, driving confidence level, etc. Such information may be utilized by the driver retirement score calculator **162** as driver retirement score manual inputs **166** for calculating the driver retirement score **163**. The driver retirement score manual inputs (i.e., self-reported inputs) **166**, **166***a*-**166***g* may include one or more of a medical condition quantity **166***a*, a medical condition type **166***b*, a complaint status **166***c*, a path following status **166***d*, a road sign compliance status **166***e*, a driver confidence status **166***f*, and a driver physical ability status **166***g*.

[0051] Additionally or alternatively, upon initiation and at **206**, the vehicle sensor system **120** begins collecting the sensor data **123**, **123***a*-**123***c* from one or more of the exterior sensor subsystem **122***a*, the interior sensor subsystem **122***b*, and the vehicle status subsystem **122***c* at a data collection step **206**. As discussed in greater detail below, the sensor data **123**, **123***a*-**123***c* may be utilized by the driver retirement score calculator **162** as driver retirement score observed inputs **168** for calculating the driver retirement score **163**. The driver retirement score observed inputs **168** may concern one or more of a driver fatigue status **168***a*, a driver distraction status **168***b*, a speed-limit following status **168***c*, a driver squinting status **168***d*, a honking status **168***e*, a turn signal utilization status **168***f*, a vehicle straddling status **168***g*, a vehicle drifting status **168***h*, a driver reaction time status **168***i*, a vehicle following distance status **168***j*, a driving record status **168***k*, a driver posture status **168***l*, a vehicle speed control status **168***m*, an accelerator and brake status **168***n*, a navigation deviation status **168***o*, a vehicle control status **168***p*, a dangerous behavior status **168***q*, and a stress or anxiety status **168***r*. Additionally, other driver retirement score observed inputs **168** not listed above can also be evaluated and provided to the driver retirement score calculator **162**, especially driver retirement score observed inputs **168** that may be desirable for determining physical or psychological decline of the driver.

[0052] At **208**, the driver retirement score **163** may be calculated using the driver evaluation module **160**. More particularly, according to at least one aspect of the present disclosure, the driver retirement score **163** may be calculated via the driver retirement score calculator **162**. The driver retirement score calculator **162** may be configured to assess an older driver's ability to drive based on the driver retirement score manual inputs **166** and/or the driver retirement score observed inputs **168** that may change over time due to physical and/or psychological progressions (hereinafter, may be referred to collectively as "the inputs **166**, **168**"). More specifically, the driver retirement score calculator **162** may be configured to aggregate the one or more inputs **166**, **168** and either increase or decrease the driver retirement score **163** accordingly. Some inputs **166**, **168** may be weighted so that they affect the driver retirement score **163** (i.e., increase or decrease the score by a greater amount). Additionally, various machine learning algorithms or techniques may be implemented to further refine the effect of each input on the driver retirement score **163**. The driver retirement score calculator **162** may also be configured to require a threshold amount of inputs **166**, **168** and/or trips recorded by the driver before providing the driver retirement score **163** to the driver assessment module **164** for further evaluation.

[0053] As indicated above, the driver retirement score manual inputs **166** may affect (e.g., increase or decrease) the calculation of the driver retirement score **163** and each input **166***a*-**166***g* can evaluate different aspects or signs commonly associated with physical and/or psychological decline of older drivers. With respect to the medical condition quantity **166***a*, the driver can voluntarily provide the quantity of known medical conditions such as those for which they have a medical diagnosis, for example. An increasing number of medical conditions may increase the driver retirement score **163**. The driver may also provide the type of medical conditions **166***b*, if any, currently faced by the driver. For example, the driver retirement score **163** may be increased if an older driver indicates that they have been diagnosed with cataracts or a cognitive disorder such as dementia or Alzheimer's disease. If the driver self-reports that they complain a lot about other drivers while driving, then the complaint status **166***c* may be affected accordingly, thereby increasing the driver retirement score **163**. The path following status **166***d* may increase the driver retirement score **163**

if the driver self-reports that they tend to have trouble with following turn-by-turn GPS guidance while driving. Similarly, the road sign compliance status **166**e may also increase the driver retirement score **163** if the driver reports that they have difficulty seeing road signs or typically fail to abide by the roads signs (e.g., fail to stop at a stop sign or fail to yield at a yield sign). Through the mobile application **32** and/or the vehicle user interface system **130**, the driver may also have the options of assessing their confidence as a driver which is reflected in the driver confidence status **166**f and also assess their physical ability which is reflected in the driver physical ability status **166**g. If the driver confidence status **166**f indicates the driver is confident in their driving ability, then the driver retirement score **163** may remain the same or decrease. Additionally, if the driver physical ability status **166**g indicates good mobility that can be associated with the necessary motor skills to drive the vehicle **100**, then the driver retirement score **163** may remain the same or decrease. But if the driver self-reports that they are not confident as a driver and/or provides that they have they have trouble moving their neck or arms, then the driver retirement score **163** may increase accordingly.

[0054] Again, as indicated above, the driver retirement score observed inputs **168** may affect (e.g., increase or decrease) the calculation of the driver retirement score **163** and each input **168**a-**168**q can evaluate different aspects or signs commonly associated with physical and/or psychological decline of older drivers. The fatigue status **168**a, the driver distraction status **168**b, the driver squinting status **168**d, and/or a driver posture status **1681** can be observed via the interior sensor system **122**b. According to at least one aspect of the present disclosure, the driver retirement score **163** may be increased if one or more cameras and/or radars provide data that indicates the driver is commonly distracted (e.g., looking at their phone), squinting, or slouching. If the speed-limit following status **168**c or the vehicle speed control status **168**m indicates that the driver doesn't follow the posted speed limit or frequently fluctuates speed, then the driver retirement score **163** may be increased or otherwise affected. The honking status **168**e considers whether the driver tends to be honked at by other drivers while driving, which may be determined by the interior or exterior sensor subsystems **122**a, **122**b via one or more microphones, for example. Thus, the driver retirement score **163** may be increased if the driver is frequently being honked at by other drivers. The exterior sensor subsystem **122**a may also help determine the vehicle straddling status **168**g, which considers how well the driver straddles a lane on a road, the drifting status **168**h, which considers if the driver tends to deviate from their path or from the lane without use of a turn signal, and the following distance status **168**j, which considers the distance maintained between the vehicle **100** and a vehicle preceding the vehicle **100**. If the driver fails to stay in their lane, drifts from their lane, or follows too close (i.e., tailgating) or too far from a vehicle preceding the vehicle **100**, then the driver retirement score **163** may increase. The turn signal utilization status **168**f concerns whether the driver uses a turn signal to change lanes or to indicate that they intend to turn left at an intersection, for example. The driver reaction time status **168**i may be determined using a combination of inputs from the vehicle subsystems **122**a-**122**c and if the driver reaction time status indicates **168**i that the driver is not stopping when they should or not proceeding through a traffic signal appropriately, then the driver retirement score **163** may be affected. The accelerator and brake status **168**n may represent a situation where the driver presses the accelerator and brake simultaneously while driving which may affect the driver retirement score **163**. The navigation deviation status **1680** may capture scenarios where the vehicle **100** (e.g., via the operation sensor subsystem **122**c) detects deviation from a determined navigation route. If the driver consistently has trouble following GPS guidance, then the driver retirement score **163** may be increased or otherwise affected. The vehicle control status **168**p may consider if the driver is an impediment to traffic by being too passive on the road (e.g., slow turning, using turn late incorrectly, coasting to a near stop amid moving traffic, etc.). If the control status **168**p indicates that the driver is failing to control the vehicle **100** properly, then the driver retirement score **100** may be affected. Consistent or habitual incidents of poor performance may be represented by the dangerous behavior status **168**q and affect the driver retirement score **163**. Through use of one or more biosensors (i.e., in the driver's seat or steering wheel) of the interior sensor subsystem **122**b, the stress and anxiety status **168**r may vary during certain driving situations that the driver is presented with and may be used to calculate the driver retirement score **163**. Additional inputs that may be used to calculate the driver retirement score **163** include situations such as the driver hitting a curb while driving, turning too sharply, parking too close to another vehicle, or running a red light, for example.

[0055] Additionally or alternatively, the driver retirement score calculator **162** may be able to access or otherwise gather information (e.g., public records) using the internet to determine pertinent information concerning a driver's performance. For example, if the driver retirement score calculator determines that the driver has received one or more citations (i.e., traffic tickets) then the driver retirement score **163** may be affected accordingly (e.g., increased).

[0056] At **210**, the driver assessment module **164** can receive the driver retirement score **163** from the driver retirement score calculator **162** and assess the driver's performance accordingly. The driver retirement score **163** may be compared to one or more of the threshold values to determine whether the driver should be assessed with the first performance category **165**a, the second performance category **165**b, or the third performance category **165**c. As indicated above, the performance categories **165**a-**165**c can be associated with acceptable driving performance, questionable driving performance, unacceptable driving performance, acceptable physical and psychological ability of the driver, questionable physical and psychological ability of the driver, and/or unacceptable physical and psychological ability of the driver. If the driver retirement score **163** is less than a first threshold value, then the driver assessment module **164** may assess the driver with the first performance category **165**a (i.e., acceptable performance category) at **214**. However, if the driver retirement score **163** is greater than the first threshold value then the method **200** can proceed to **212**.

[0057] At **212**, if the driver retirement score **163** is greater than the first threshold value, but less than a second threshold value, then the driver assessment module **164** may assess the driver with the second performance category (i.e., questionable performance) at **216**. If, however, the driver retirement score **163** is greater than both the first and second thresholds, then the driver assessment module **164** may assess the driver with the third performance category (i.e., unacceptable performance) at **218**.

[0058] At **220**, if this was the first time the driver was assessed with one of the performance categories **165***a*-**165***c*, then a report **169** (FIG. **2**) will be generated at **222**. The report can include personalized (i.e., tailored) recommendations based on the inputs **166**, **168** and/or the performance category **165***a*-**165***c* in which the driver falls. For instance, the recommendations may include possible options for avoiding scenarios where the driver may struggle. If the driver retirement score indicates poor performance while driving during the night time or while driving during harsh weather conditions (e.g., heavy rain or snow), then one of the recommendations on the report **169** may be to avoid driving in such scenarios. Thus, the report **169** may include one or more restrictions or limitations concerning what time of day to drive, the road conditions to drive in, the type of car to drive, and/or whether an autonomous driving feature should be enabled.

[0059] If the driver assessment at **220** is not the first time the driver is being assessed with one of the performance categories **165***a*-**165***c*, then the method **200** proceeds to **224** where the performance category is compared against prior assessments of the driver. In other words, the driver performance category **165** can be refined over time and monitored for degradation or improvement. For instance, if there was an improvement of driver performance between a prior assessment and the driver's most recent assessment then that is considered when generating the report **169** at **222**. Thus, if the driver was assessed with the third performance category **165***c* when previously evaluated while driving at night (i.e., low visibility), but is currently being assessed with the first or the second category **165***a*, **165***b* while driving during the daytime, then the report may provide personalized recommendations suggesting that the driver should only drive during the day if possible. However, if the driver has been assessed more than once and there is not much variation among the assessments, for example, the driver has been assessed with the third performance category **165***c* several times, then the assessment report **169** may begin providing recommendations suggesting that the driver should avoid driving as much as possible and also present alternatives for traveling on the road (e.g., get a ride from a family member or a friend).

[0060] At **226**, the driver may be notified of the report **169** via the vehicle user interface system **130** and/or the mobile device **30** (i.e., through the mobile application **32**). Additionally or alternatively, a designated contact of the driver may be notified of the report **169** on a mobile device or personal computer via the network **40**.

[0061] At **228**, the method **200** ends. Alternatively, as shown by dashed lines, the method **200** may return to **206** from **224** or **226** and the performance of the driver may be evaluated and assessed again.

[0062] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

[0063] The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A computer-implemented method that, when executed by data processing hardware, causes the data processing hardware to perform operations comprising:

gathering vehicle and driver data;

receiving one or more scoring inputs based on the vehicle and driver data;

calculating a driver retirement score based on the one or more scoring inputs;

assessing a driver of a vehicle with a performance category based on the driver retirement score; and

notifying the driver of the performance category.

2. The method of claim **1**, wherein the one or more scoring inputs comprises driver retirement score manual inputs and driver retirement score observed inputs.

3. The method of claim **1**, wherein the one or more scoring inputs comprises driver retirement score manual inputs that increase or decrease the driver retirement score and include at least one of the following:

a medical condition quantity;

a medical condition type;

a complaint status;

a path following status;

a road sign compliance status;

a driver confidence status; and

a driver physical ability status.

4. The method of claim **1**, wherein the one or more scoring inputs comprises driver retirement score observed inputs that increase or decrease the driver retirement score and include at least one of the following:

a driver fatigue status;

a driver distraction status;

a speed-limit following status;

a driver squinting status;

a honking status;

a turn signal utilization status;

a vehicle straddling status;

a vehicle drifting status;

a driver reaction time status;

a vehicle following distance status;

a driving record status;

a driver posture status;

a vehicle speed control status;

an accelerator and brake status;

a navigation deviation status;

a vehicle control status; and

a dangerous behavior status.

5. The method of claim **1**, wherein the performance category comprises a first performance category, a second performance category, and a third performance category.

6. The method of claim **5**, wherein the first performance category is associated with acceptable driver performance, the second performance category is associated with questionable driver performance, and the third performance category is associated with unacceptable driver performance.

7. The method of claim **5**, wherein the first performance category is associated with acceptable physical and psychological ability for driving, the second performance category is associated with questionable physical and psychological ability for driving, and the third performance category is associated with unacceptable physical and psychological ability for driving.

8. The method of claim **1**, wherein assessing the driver of the vehicle with the performance category based on the driver retirement score comprises:

assessing a first performance category if the driver retirement score is less than a first threshold;

assessing a second performance category if the driver retirement score is greater than the first threshold but less than a second threshold; and

assessing a third performance category if the driver retirement score is greater than both the first and second thresholds.

9. The method of claim **1**, further comprising generating a report based on the one or more scoring inputs and the performance category, and providing personalized recommendations for the driver via a user interface system or a mobile device.

10. The method of claim **1**, further comprising determining whether the performance of the driver is improving or degrading and providing personalized recommendations in a report based on any improvement or degradation of performance.

11. The method of claim **1**, wherein notifying the driver further comprises notifying a designated contact of the driver.

12. A vehicle management system comprising:

a sensor system comprising one or more sensor subsystems;

data processing hardware; and

memory hardware in communication with the data processing hardware, the memory hardware storing instructions that, when executed on the data processing hardware, cause the data processing hardware to perform operations comprising:

gathering vehicle and driver data via the sensor system and via one or more manual inputs from a driver;

receiving one or more scoring inputs based on the vehicle and driver data;

calculating a driver retirement score based on the one or more scoring inputs;

assessing a driver of a vehicle with a performance category based on the driver retirement score;

generating a report with personalized recommendations for the driver based on the one or more inputs and the performance category; and

notifying the driver and one or more designated contacts of the report.

13. The system of claim **12**, wherein the one or more scoring inputs comprises driver retirement score manual inputs and driver retirement score observed inputs.

14. The system of claim **12**, wherein the one or more scoring inputs comprises driver retirement score manual inputs that increase or decrease the driver retirement score and include at least one of the following:

a medical condition quantity;

a medical condition type;

a complaint status;

a path following status;

a road sign compliance status;

a driver confidence status; and

a driver physical ability status.

15. The system of claim **12**, wherein the one or more scoring inputs comprises driver retirement score observed inputs that increase or decrease the driver retirement score and include at least one of the following:

a driver fatigue status;

a driver distraction status;

a speed-limit following status;

a driver squinting status;

a honking status;

a turn signal utilization status;

a vehicle straddling status;

a vehicle drifting status;

a driver reaction time status;

a vehicle following distance status;

a driving record status;

a driver posture status;

a vehicle speed control status;

an accelerator and brake status;

a navigation deviation status;

a vehicle control status; and

a dangerous behavior status.

16. The system of claim **12**, wherein the performance category comprises a first performance category, a second performance category, and a third performance category.

17. The system of claim **16**, wherein the first performance category is associated with acceptable driver performance, the second performance category is associated with questionable driver performance, and the third performance category is associated with unacceptable driver performance.

18. The system of claim **16**, wherein the first performance category is associated with acceptable physical and psychological ability for driving, the second performance category is associated with questionable physical and psychological ability for driving, and the third performance category is associated with unacceptable physical and psychological ability for driving.

19. The system of claim **12**, wherein assessing the driver of the vehicle with the performance category based on the driver retirement score comprises:

assessing a first performance category if the driver retirement score is less than a first threshold;

assessing a second performance category if the driver retirement score is greater than the first threshold but less than a second threshold; and

assessing a third performance category if the driver retirement score is greater than both the first and second thresholds.

20. The system of claim **12**, further comprising determining whether the performance of the driver is improving or degrading and providing personalized recommendation in the report based on any improvement or degradation of performance.

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